

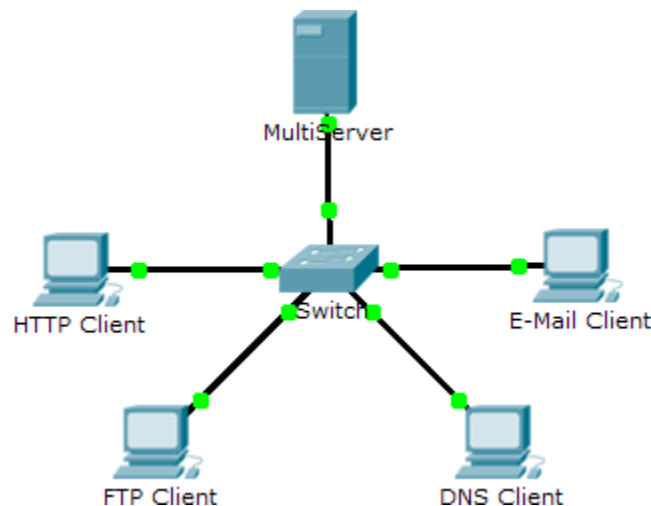
Packet Tracer Simulation - TCP and UDP Communications

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Section: 12

Topology



Objectives

Part 1: Generate Network Traffic in Simulation Mode

Part 2: Examine the Functionality of the TCP and UDP Protocols

Background

This simulation activity is intended to provide a foundation for understanding the TCP and UDP in detail. Simulation mode provides the ability to view the functionality of the different protocols.

As data moves through the network, it is broken down into smaller pieces and identified in some fashion so that the pieces can be put back together. Each of these pieces is assigned a specific name (protocol data unit [PDU]) and associated with a specific layer. Packet Tracer Simulation mode enables the user to view each of the protocols and the associated PDU. The steps outlined below lead the user through the process of requesting services using various applications available on a client PC.

This activity provides an opportunity to explore the functionality of the TCP and UDP protocols, multiplexing and the function of port numbers in determining which local application requested the data or is sending the data.

Part 1: Generate Network Traffic in Simulation Mode

Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

Perform the following tasks to reduce the amount of network traffic viewed in the simulation.

- Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.
- Enter the **ping 192.168.1.255** command. This will take a few seconds as every device on the network responds to **MultiServer**.
- Close the **MultiServer** window.

Step 2: Generate web (HTTP) traffic.

- a. Switch to Simulation mode.
- b. Click **HTTP Client** and click the **Desktop** tab > **Web Browser**.
- c. In the URL field, enter **192.168.1.254** and click **Go**. Envelopes (PDUs) will appear in the simulation window.
- d. Minimize, but do not close, the **HTTP Client** configuration window.

Step 3: Generate FTP traffic.

- a. Click **FTP Client** and click the **Desktop** tab > **Command Prompt**.
- b. Enter the **ftp 192.168.1.254** command. PDUs will appear in the simulation window.
- c. Minimize, but do not close, the **FTP Client** configuration window.

Step 4: Generate DNS traffic.

- a. Click **DNS Client** and click the **Desktop** tab > **Command Prompt**.
- b. Enter the **nslookup multiserver.pt.ptu** command. A PDU will appear in the simulation window.
- c. Minimize, but do not close, the **DNS Client** configuration window.

Step 5: Generate Email traffic.

- a. Click **E-Mail Client** and click the **Desktop** tab > **E Mail** tool.
- b. Click **Compose** and enter the following information:
 - 1) **To:** user@multiserver.pt.ptu
 - 2) **Subject:** Personalize the subject line
 - 3) **E-Mail Body:** Personalize the Email
- c. Click **Send**.
- d. Minimize, but do not close, the **E-Mail Client** configuration window.

Step 6: Verify that the traffic is generated and ready for simulation.

Every client computer should have PDUs listed in the Simulation Panel.

Part 2: Examine Functionality of the TCP and UDP Protocols

Step 1: Examine multiplexing as all of the traffic crosses the network.

You will now use the **Capture/Forward** button and the **Back** button in the Simulation Panel.

- a. Click **Capture/Forward** once. All of the PDUs are transferred to the switch.
- b. Click **Capture/Forward** again. Some of the PDUs disappear. What do you think happened to them?

[PDUs are stored in the switch.](#)

- c. Click **Capture/Forward** six times. All clients should have received a reply. Note that only one PDU can cross a wire in each direction at any given time. What is this called?

[Multiplexing](#)

- d. A variety of PDUs appears in the event list in the upper right pane of the simulation window. Why are they so many different colors?

To represent different protocols.

- e. Click **Back** eight times. This should reset the simulation.

Note: Do not click **Reset Simulation** any time during this activity; if you do, you will need to repeat the steps in Part 1.

Step 2: Examine HTTP traffic as the clients communicate with the server.

- Filter the traffic that is currently displayed to display only **HTTP** and **TCP** PDUs filter the traffic that is currently displayed:
 - Click **Edit Filters** and toggle the **Show All/None** check box.
 - Select **HTTP** and **TCP**. Click anywhere outside of the Edit Filters box to hide it. The Visible Events should now display only **HTTP** and **TCP** PDUs.
- Click **Capture/Forward**. Hold your mouse above each PDU until you find one that originates from **HTTP Client**. Click the PDU envelope to open it.
- Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

TCP

Are these communications considered to be reliable?

Yes, it is considered to be reliable.

- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

SOURCE PORT:1025

DEST PORT: 80

SEQUENCE NUM: 0

ACK NUM: 0

SYN is written in the field to the left of the WINDOW field because FLAGS is 0b00000010

TCP																Bits															
0						4						8						16						24						Bits	
SOURCE PORT:1025																DESTINATION PORT:80															
SEQUENCE NUMBER:0																															
ACKNOWLEDGEMENT NUMBER:0																												SEQUENCE NUMBER:0			
OFFSET:0x0				RESERVED: 0				FLAGS:0b00000010								WINDOW:65535															
CHECKSUM:0x0000																URGENT POINTER:0x0000															
OPTION																															
DATA (VARIABLE LENGTH)																								PADDING: 0							

Packet Tracer Simulation - TCP and UDP Communications

- e. Close the PDU and click **Capture/Forward** until a PDU returns to the **HTTP Client** with a checkmark.
- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

SOURCE PORT: 80

DEST PORT: 1025

SEQUENCE NUM: 0

ACK NUM: 1

The source port number and destination port number are swapped. Acknowledgement number becomes 1, and SYN changed to SYN+ACK due to FLAGS changes to 0b00010010.

TCP																								Bits
0 4 8 16 24																								
SOURCE PORT:80												DESTINATION PORT:1025												
SEQUENCE NUMBER:0																								
ACKNOWLEDGEMENT NUMBER:1																								
OFFSET:0x0				RESERVED: 0				FLAGS:0b00010010								WINDOW:16384								
CHECKSUM:0x0000												URGENT POINTER:0x0000												
OPTION																								
DATA (VARIABLE LENGTH)																		PADDING: 0						

- g. There is a second **PDU** of a different color, which **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the HTTP communication. Click this second PDU envelope and select **Outbound PDU Details**.
- h. What information is now listed in the TCP section? How are the port and sequence numbers different from the previous two PDUs?

SOURCE PORT:1025

DEST PORT: 80

SEQUENCE NUM: 1

ACK NUM: 1

The source port number and destination port number are the same with first PDU, while reversed in the second PDU. Sequence number and acknowledgement number become 1, and PSH+SYN is written due to FLAGS changes to 0b00011000.

TCP																Bits																															
0								4																8								16								24							
SOURCE PORT:1025																								DESTINATION PORT:80																							
SEQUENCE NUMBER:1																																															
ACKNOWLEDGEMENT NUMBER:1																																															
OFFSET:0x0				RESERVED: 0				FLAGS:0b00011000								WINDOW:65535																															
CHECKSUM:0x0000																URGENT POINTER:0x0000																															
OPTION																																															
DATA (VARIABLE LENGTH)																								PADDING: 0																							

- i. Click **Back** until the simulation is reset.

Step 3: Examine FTP traffic as the clients communicate with the server.

- In the Simulation Panel, change **Edit Filters** to display only **FTP** and **TCP**.
- Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **FTP Client**. Click that PDU envelope to open it.
- Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

TCP

Are these communications considered to be reliable?

Yes, it is reliable

- Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

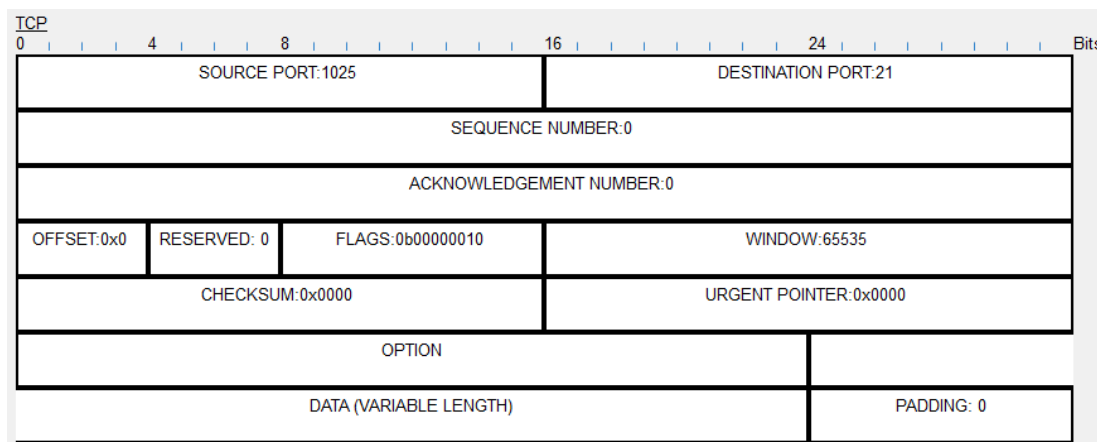
SOURCE PORT:1025

DEST PORT: 21

SEQUENCE NUM: 0

ACK NUM: 0

SYN is written in the field to the left of the WINDOW field because FLAGS is 0b00000010



- Close the PDU and click **Capture/Forward** until a PDU returns to the **FTP Client** with a checkmark.

- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

SOURCE PORT: 21

DEST PORT: 1025

SEQUENCE NUM: 0

ACK NUM: 1

The source port number and destination port number are swapped. Acknowledgement number becomes 1, and SYN changed to SYN+ACK due to FLAGS changes to 0b00010010.

TCP																								Bits
0				4				8				16				24								
SOURCE PORT:21												DESTINATION PORT:1025												
SEQUENCE NUMBER:0																								
ACKNOWLEDGEMENT NUMBER:1																								
OFFSET:0x0				RESERVED: 0				FLAGS:0b00010010								WINDOW:16384								
CHECKSUM:0x0000												URGENT POINTER:0x0000												
OPTION																								
DATA (VARIABLE LENGTH)																		PADDING: 0						

- g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?

SOURCE PORT:1025

DEST PORT: 21

SEQUENCE NUM: 1

ACK NUM: 1

The source port number and destination port number are the same with first PDU, while reversed in the second PDU. Sequence number and acknowledgement number become 1, and SYN is changed to ACK due to FLAGS changes to 0b00010000.

TCP

0481624

Bits

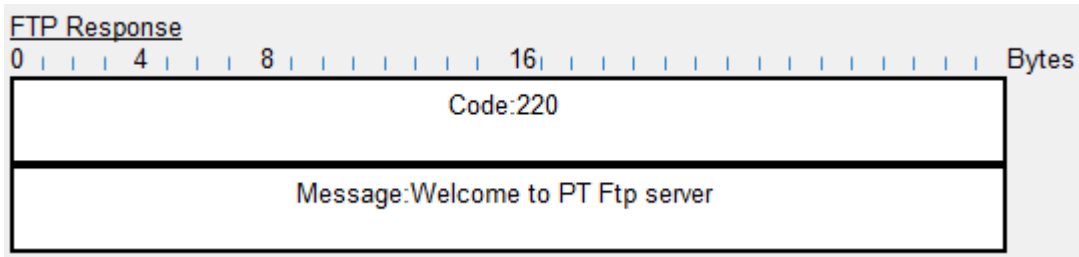
SOURCE PORT:1025																DESTINATION PORT:21															
SEQUENCE NUMBER:1																															
ACKNOWLEDGEMENT NUMBER:1																															
OFFSET:0x0				RESERVED: 0				FLAGS:0b00010000												WINDOW:65535											
CHECKSUM:0x0000																URGENT POINTER:0x0000															
OPTION																															
DATA (VARIABLE LENGTH)																								PADDING: 0							

- h. Close the PDU and click **Capture/Forward** until a second PDU returns to the **FTP Client**. The PDU is a different color.

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- i. Open the PDU and select **Inbound PDU Details**. Scroll down past the TCP section. What is the message from the server?

Welcome to PT Ftp server



- j. Click **Back** until the simulation is reset.

Step 4: Examine DNS traffic as the clients communicate with the server.

- a. In the Simulation Panel, change **Edit Filters** to display only **DNS** and **UDP**.
- b. Click the PDU envelope to open it.
- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?
UDP

Are these communications considered to be reliable?

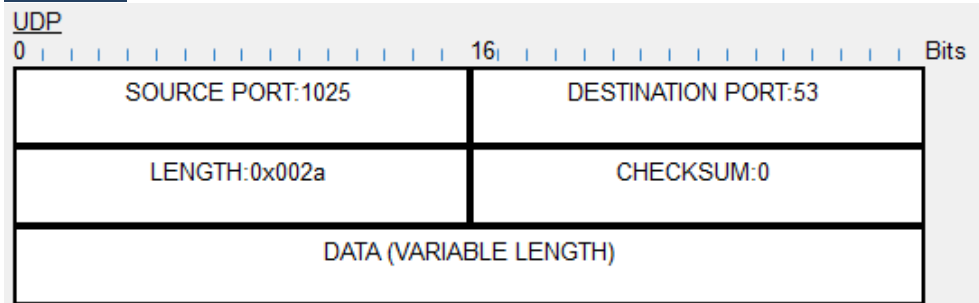
No, it is not reliable

- d. Record the **SRC PORT** and **DEST PORT** values. Why is there no sequence and acknowledgement number?

SOURCE PORT:1025

DEST PORT: 53

There are no sequence and acknowledgement number because UDP does not need to build reliable connection



- e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **DNS Client** with a checkmark.

Packet Tracer Simulation - TCP and UDP Communications

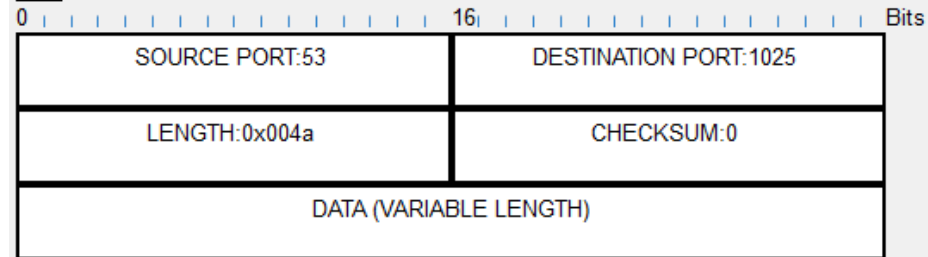
- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

SOURCE PORT:53

DEST PORT: 1025

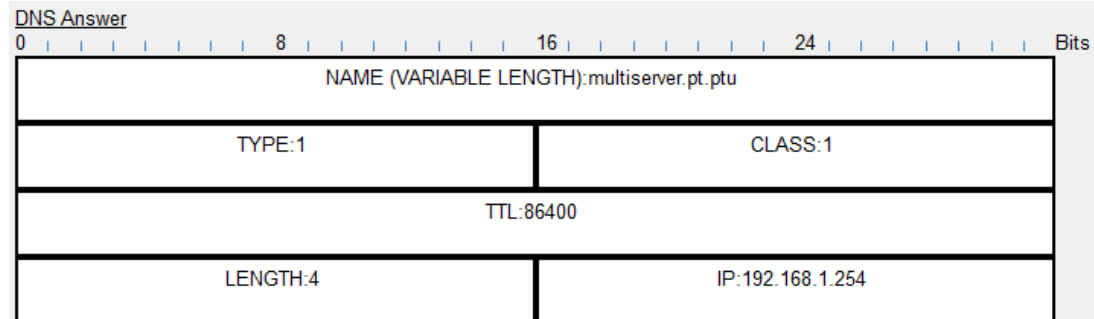
The source port number and destination port number are swapped

UDP



- g. What is the last section of the **PDU** called?

DNS Answer



- h. Click **Back** until the simulation is reset.

Step 5: Examine email traffic as the clients communicate with the server.

- In the Simulation Panel, change **Edit Filters** to display only **POP3, SMTP** and **TCP**.
- Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **E-mail Client**. Click that PDU envelope to open it.
- Click the **Inbound PDU Details** tab and scroll down to the last section. What transport layer protocol does email traffic use?

TCP

Are these communications considered to be reliable?

Yes, it is reliable

- Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

SOURCE PORT:1025

DEST PORT: 25

SEQUENCE NUM: 0

ACK NUM: 0

SYN is written in the field to the left of the WINDOW field because FLAGS is 0b00000010

TCP																Bits																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
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- Close the **PDU** and click **Capture/Forward** until a PDU returns to the **E-Mail Client** with a checkmark.

- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

SOURCE PORT: 25

DEST PORT: 1025

SEQUENCE NUM: 0

ACK NUM: 1

The source port number and destination port number are swapped. Acknowledgement number becomes 1, and SYN changed to SYN+ACK due to FLAGS changes to 0b00010010.

TCP																Bits																			
0				4				8				16				24																			
SOURCE PORT:25												DESTINATION PORT:1025																							
SEQUENCE NUMBER:0																																			
ACKNOWLEDGEMENT NUMBER:1																																			
OFFSET:0x0				RESERVED: 0				FLAGS:0b00010010								WINDOW:16384																			
CHECKSUM:0x0000																URGENT POINTER:0x0000																			
OPTION																																			
DATA (VARIABLE LENGTH)																								PADDING: 0											

- g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?

SOURCE PORT:1025

DEST PORT: 25

SEQUENCE NUM: 1

ACK NUM: 1

The source port number and destination port number are the same with first PDU, while reversed in the second PDU. Sequence number and acknowledgement number become 1, and SYN is changed to ACK due to FLAGS changes to 0b00010000.

TCP																								Bits							
0				4				8				16				24															
SOURCE PORT:1025												DESTINATION PORT:25																			
SEQUENCE NUMBER:1																															
ACKNOWLEDGEMENT NUMBER:1																															
OFFSET:0x0				RESERVED: 0				FLAGS:0b00010000								WINDOW:65535															
CHECKSUM:0x0000												URGENT POINTER:0x0000																			
OPTION																															
DATA (VARIABLE LENGTH)																		PADDING: 0													

- h. There is a second **PDU** of a different color that **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the email communication. Click this second PDU envelope and select **Outbound PDU Details**.

Packet Tracer Simulation - TCP and UDP Communications

- i. How are the port and sequence numbers different from the previous two PDUs?

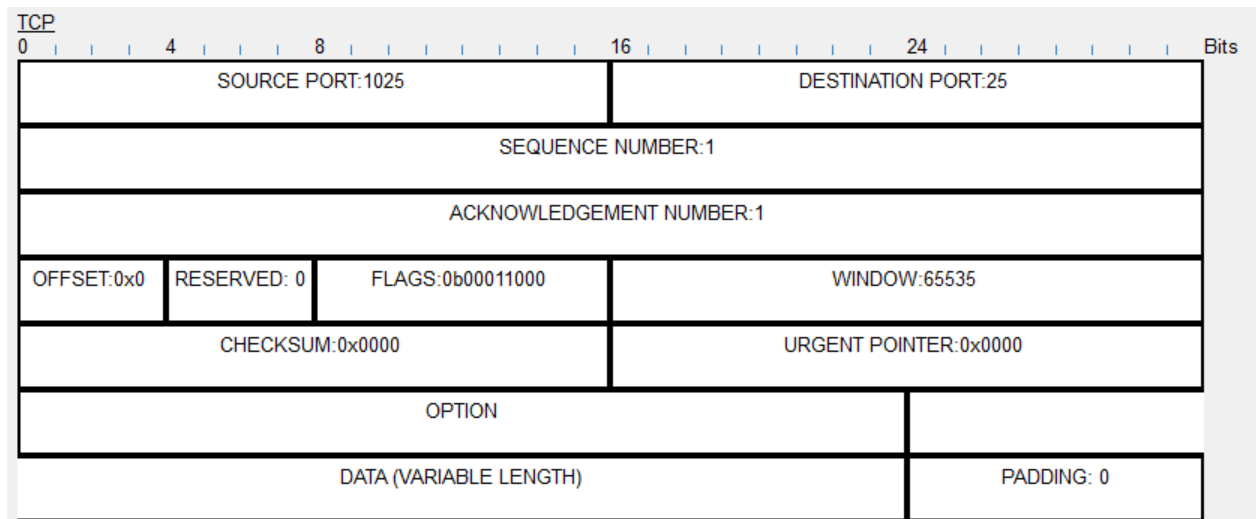
SOURCE PORT:1025

DEST PORT: 25

SEQUENCE NUM: 1

ACK NUM: 1

The source port number and destination port number are the same with first PDU, while reversed in the second PDU. Sequence number and acknowledgement number become 1. SYN and SYN+ACK changed to PSH+ACK due to FLAGS changes to 0b00011000.



- j. What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?

SMTP is associated with TCP port 25 and POP3 is associated with TCP port 110.

- k. Click **Back** until the simulation is reset.

Step 6: Examine the use of port numbers from the server.

- a. To see TCP active sessions, perform the following steps in quick succession:

- 1) Switch back to **Realtime** mode.
- 2) Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.

- b. Enter the **netstat** command. What protocols are listed in the left column?

TCP

- c. What port numbers are being used by the server?

The port number is 21.

```
Active Connections

  Proto Local Address           Foreign Address         State
  TCP    192.168.1.254:21        192.168.1.2:1025      ESTABLISHED
C:\>
```

- d. What states are the sessions in?

ESTABLISHED

- e. Repeat the **netstat** command several times until you see only one session still ESTABLISHED. For which service is this connection still open?

FTP Client

```
C:\>netstat

Active Connections

  Proto Local Address           Foreign Address         State
  TCP    192.168.1.254:21        192.168.1.2:1025      ESTABLISHED
C:\>netstat

Active Connections

  Proto Local Address           Foreign Address         State
  TCP    192.168.1.254:21        192.168.1.2:1025      ESTABLISHED
C:\>netstat

Active Connections

  Proto Local Address           Foreign Address         State
  TCP    192.168.1.254:21        192.168.1.2:1025      ESTABLISHED
C:\>netstat

Active Connections

  Proto Local Address           Foreign Address         State
  TCP    192.168.1.254:21        192.168.1.2:1025      ESTABLISHED
C:\>netstat

Active Connections

  Proto Local Address           Foreign Address         State
  TCP    192.168.1.254:21        192.168.1.2:1025      ESTABLISHED
C:\>
```

Why doesn't this session close like the other three? (Hint: Check the minimized clients)
[The server is waiting for a username from the client.](#)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 192.168.1.254
Trying to connect...192.168.1.254
Connected to 192.168.1.254
220- Welcome to PT Ftp server
Username:
```

Suggested Scoring Rubric

Activity Section	Question Location	Possible Points	Earned Points
Part 2: Examine Functionality of the TCP and UDP Protocols	Step 1	15	
	Step 2	15	
	Step 3	15	
	Step 4	15	
	Step 5	15	
	Step 6	25	
Total Score		100	